PRACTICE SHEET

1. If the matrix $\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is singular, then what is one of the values of θ?
(a) $\pi/4$ (b) $\pi/2$
(c) π (d) 0

2. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix}$, If AB = BA , then what is the value of x?
(a) −1 (b) 0
(c) 1 (d) Any real number

3. Let $A = (a_{ij})_{n\times n}$ and $\text{adj}(A) = (\alpha_{ij})$, If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 2 & 3 & -1 \end{bmatrix}$ then what is the value of α_{23} ?
(a) 1 (b) −1
(c) 8 (d) −8

4. Let A and B be two invertible matrices each of order n . What is adj(AB) equal to ?
(a) (adjA) (adjB) (b) (adjA) + (adjB)
(c) (adjA) − (adjB) (d) (adjB) (adjA)

5. M is a matrix with real entries given by
$M = \begin{bmatrix} 4 & k & 0 \\ 6 & 3 & 0 \\ 2 & t & k \end{bmatrix}$, which of the following conditions guarantee the invertibility of M?
1. $k \neq 2$ 2. $k \neq 1$
3. $t \neq 0$ 4. $t \neq 1$
Select the correct answer using the code given below:
(a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 3 and 4

6. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ be a square matrix of order 3. Then for any positive integer n, what is A^n equal to?
(a) A (b) 3^n A
(c) (3^{n-1}) A (d) 3A

7. If A is a square matrix such that $A - A^T = 0$, then which one of the following is correct?
(a) A must be a null matrix
(b) A must be a unit matrix
(c) A must be a scalar matrix
(d) None of the above

8. What is the inverse of $A = \begin{bmatrix} 1+i & 1+i \\ -1+i & 1-i \end{bmatrix}$?
(a) $\frac{1}{4}\begin{bmatrix} 1-i & -1-i \\ 1-i & 1-i \end{bmatrix}$ (b) $\frac{1}{4}\begin{bmatrix} 1+i & -1+i \\ 1+i & -1-i \end{bmatrix}$
(c) $\frac{1}{4}\begin{bmatrix} 1+i & 1-i \\ -1-i & 1+i \end{bmatrix}$ (d) $\frac{1}{4}\begin{bmatrix} 1+i & 1-i \\ -1-i & -1+i \end{bmatrix}$

9. Let A be an m × n matrix. Under which one of the following conditions does A^{-1} exists?
(a) m = n only
(b) m = n and $\det(A) \neq 0$
(c) m = n and det(A) = 0
(d) $m \neq n$

10. Let A and B be two matrices of order n×n. Let A be a non singular. Consider the following:
(a) AB is a singular
(b) AB is a non singular
(c) $A^{-1}B$ is a singular
(d) $A^{-1}B$ is non singular

11. The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$ satisfies which one of the following polynomial equations?
(a) $A^2 + 3A + 2I = 0$ (b) $A^2 + 3A - 2I = 0$
(c) $A^2 - 3A - 2I = 0$ (d) $A^2 - 3A + 2I = 0$

12. If a matrix A is symmetric as well as anti-symmetric, then which one of the following is correct?
(a) A is a diagonal matrix
(b) A is a null matrix
(c) A is a unit matrix
(d) A is a triangular matrix

13. If $A = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, where ω is cube root of unity, then what is A^{100} equal to:
(a) A (b) −A
(c) Null matrix (d) Identity matrix

14. If a matrix X has (a+b) rows and (a+2) columns and a matrix Y has (b+1) rows and (a+3) columns. If both XY and YX exist, then what are the values of a, b respectively?
(a) 3,2 (b) 2, 3
(c) 2,4 (d) 4,3

15. If X and Y are the matrices of order 2 × 2 each and 2x−3y $= \begin{bmatrix} -7 & 0 \\ 7 & -13 \end{bmatrix}$ and $3x + 2y = \begin{bmatrix} 9 & 13 \\ 4 & 13 \end{bmatrix}$, then what is Y equal to:
(a) $\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 3 & 2 \\ -1 & 5 \end{bmatrix}$ (d) $\begin{bmatrix} 3 & 2 \\ 1 & -5 \end{bmatrix}$

16. If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, then A. adj A is equal to:
(a) $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 10 \\ 10 & 0 \end{bmatrix}$
(c) $\begin{bmatrix} 10 & 10 \\ 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 10 & 10 \end{bmatrix}$

17. If A= $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$ and B = $\begin{bmatrix}a & 0\\0 & b\end{bmatrix}$ where a, b are natural numbers, then which one of the following is correct?
(a) There exist more than one but finite number of B's such that AB=BA
(b) There exists exactly one B such that AB = BA
(c) There exists infinitely B's such that AB = BA
(d) There cannot exist any B such that AB = BA

18. What is the inverse of $\begin{bmatrix}0 & 0 & 1\\0 & 1 & 0\\1 & 0 & 0\end{bmatrix}$?

(a) $\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$ (b) $\begin{bmatrix}0 & 0 & 1\\0 & 1 & 0\\1 & 0 & 0\end{bmatrix}$

(c) $\begin{bmatrix}-1 & 0 & 1\\0 & -1 & 0\\0 & 0 & -1\end{bmatrix}$ (d) $\begin{bmatrix}0 & 0 & 1\\0 & -1 & 0\\-1 & 0 & 0\end{bmatrix}$

19. Consider the following statements in respect of symmetric matrices A and B.
I.AB is symmetric
II.$A^2 + B^2$ is symmetric
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

20. Which one of the following is correct is respect of the matrix?
A= $\begin{bmatrix}0 & 0 & -1\\0 & -1 & 0\\-1 & 0 & 0\end{bmatrix}$?
(a) A^{-1} does not exist (b) A = (–1)I
(c) A is a unit matrix (d) $A^2 = I$

21. Let A and B matrices of order 3 × 3. If AB = 0, then which of the following can be concluded?
(a) A = 0 and B = 0
(b)|A|=0 and |B|=0
(c) Either |A|=0 or |B|=0
(d) Either A = 0 and B = 0

22. Consider the following statements in respect of a square matrix A and its transpose A^T.
I.A + A^T is always symmetric
II.A – A^T is always anti-symmetric
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

23. If A= $\begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}$, B = $\begin{bmatrix}i & 0\\0 & -i\end{bmatrix}$, C = $\begin{bmatrix}0 & -i\\-i & 0\end{bmatrix}$, then which one of the following is not correct?
(a) $A^2 = B^2$ (b) $B^2 = C^2$
(c) AB = C (d) AB = BA

24. If A is a square matrix, then what is adj (A^T)–$(\text{adj } A)^T$ equal to?
(a) 2|A| (b) 2|A|I
(c) Null matrix (d) Unit matrix

25. If a matrix A is such that $3A^3 + 2A^2 + 5A + I = 0$, then A^{-1} is equal to:
(a) $-(3A^2 + 2A + 5)$ (b) $3A^2 + 2A + 5$
(c) $3A^2 - 2A - 5$ (d) None of these

26. Let A = $\begin{bmatrix}5 & 6 & 1\\2 & -1 & 5\end{bmatrix}$, if there exists a matrix B such that AB= $\begin{bmatrix}35 & 49\\29 & 13\end{bmatrix}$, then what is B equal to?

(a) $\begin{bmatrix}5 & 1 & 4\\2 & 6 & 3\end{bmatrix}$ (b) $\begin{bmatrix}2 & 6 & 3\\5 & 1 & 4\end{bmatrix}$

(c) $\begin{bmatrix}5 & 2\\1 & 6\\4 & 3\end{bmatrix}$ (d) $\begin{bmatrix}2 & 5\\6 & 1\\3 & 4\end{bmatrix}$

27. Consider the following statements:
I. If A'=A, then A is singular matrix, where A' is the transpose of A.
II. If A is a square matrix such that $A^3 = I$, then A is non-singular.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

28. Let A= $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$, $[a_{ij}]$, where, i, j = 1,2. If its inverse matrix is $[b_{ij}]$, then what is b_{22} equal to?
(a) –2 (b) 1
(c) 1/3 (d) –1/2

ANSWER KEY																			
1.	a	**2.**	b	**3.**	c	**4.**	d	**5.**	a	**6.**	c	**7.**	d	**8.**	a	**9.**	b	**10.**	b
11.	C	**12.**	b	**13.**	a	**14.**	b	**15.**	c	**16.**	a	**17.**	c	**18.**	b	**19.**	b	**20.**	d
21.	c	**22.**	c	**23.**	b	**24.**	c	**25.**	a	**26.**	c	**27.**	b	**28.**	d				

Solutions

Sol.1. (a)
$$\begin{vmatrix}\cos\theta & \sin\theta\\ \sin\theta & \cos\theta\end{vmatrix} = 0$$
$\cos^2\theta - \sin^2\theta = 0$
$\cos 2\theta = 0$
$\theta = \dfrac{\pi}{4}$

Sol.2. (b)
AB = BA
$$\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}1 & x\\0 & 1\end{bmatrix} = \begin{bmatrix}1 & x\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}1 & x\\ 0 & -1\end{bmatrix} = \begin{bmatrix}1 & -x\\ 0 & -1\end{bmatrix} \Rightarrow x = 0$$

Sol.3. (c)

$$\alpha_{23} = a_{32} = (-1)^{2+3}$$

$$\begin{vmatrix}1 & 3\\ 4 & 4\end{vmatrix} = -(4-12) = 8$$

Sol.4. (d)

Since A and B be two invertible square matrices each of order n then $(AB)^{-1} = (B^{-1})(A^{-1})$

$$\Rightarrow \frac{adj(AB)}{|AB|} = \frac{adj(B)}{|B|}\cdot\frac{adj(A)}{|A|}$$

Since |AB| = |B| |A|

then adj(AB) = (adjB)(adjA)

Sol.5. (a)

$$M = \begin{vmatrix}4 & k & 0\\ 6 & 3 & 0\\ 2 & t & k\end{vmatrix} \neq 0$$

$\Rightarrow k(12-6k) \neq 0$

$\Rightarrow k \neq 0$ and $k \neq 2$

Sol.6. (c)

$$A = \begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

$$A^2 = \begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} = \begin{bmatrix}3 & 3 & 3\\ 3 & 3 & 3\\ 3 & 3 & 3\end{bmatrix} = 3\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

$$A^3 = 3\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} = 3\begin{bmatrix}3 & 3 & 3\\ 3 & 3 & 3\\ 3 & 3 & 3\end{bmatrix}$$

$$= 9\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} = 3^2A$$

Hence $A^n = 3^{n-1} A$

Sol.7. (d)

$A = A^T$ answer is symmetric matrix

Sol.8. (a)

$$A = \begin{bmatrix}1+i & 1+i\\ -1+i & 1-i\end{bmatrix}$$

$$adj(A) = \begin{bmatrix}1-i & -1-i\\ 1-i & 1+i\end{bmatrix} \text{ and } |A| = 4$$

$$\text{then } A^{-1} = \frac{1}{4}\begin{bmatrix}1-i & -1-i\\ 1-i & 1-i\end{bmatrix}$$

Sol.9. (b)

Sol.10. (b)

|AB| = |A||B|

Sol.11. (c)

$$A = \begin{bmatrix}1 & 2\\ 2 & 2\end{bmatrix}$$

$$A^2 = \begin{bmatrix}1 & 2\\ 2 & 2\end{bmatrix}\begin{bmatrix}1 & 2\\ 2 & 2\end{bmatrix} = \begin{bmatrix}5 & 6\\ 6 & 8\end{bmatrix}$$

Let $A^2 + xA + yI = 0$ where x and y are constants.

$A^2 + xA + yI = 0$

$$\begin{bmatrix}5 & 6\\ 6 & 8\end{bmatrix} + x\begin{bmatrix}1 & 2\\ 2 & 2\end{bmatrix} + y\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix} = 0$$

$$\begin{bmatrix}5+x+y & 6+2x\\ 6+2x & 8+2x+y\end{bmatrix} = 0$$

by solving it x = −3 and y = −2

so answer is 3rd option.

Sol.12. (b)

Since A' = A and A' = −A

⇒A = − A

⇒A = O

Sol.13. (a)

Sol.14. (b)

The order of given matrices are

$[X]_{(a+b)\times(a+2)}$ and $[Y]_{(b+1)\times(a+3)}$

As [XY] and [YX] both exist

$\therefore$ a+2 = b + 1

and a + 3 = a+ b

⇒a = 2, b = 3

Sol.15. (c)

$$\text{Given, } 2x-3y = \begin{bmatrix}-7 & 0\\ 7 & -13\end{bmatrix} \quad \ldots(i)$$

$$\text{And } 3x+2y = \begin{bmatrix}9 & 13\\ 4 & 13\end{bmatrix} \quad \ldots(ii)$$

On multiplying Eq. (i) by 3 and Eq. (ii) by 2 and subtracting Eq.(i) from Eq. (ii) we, get

$$13Y = 2\begin{bmatrix}9 & 13\\ 4 & 13\end{bmatrix} - 3\begin{bmatrix}-7 & 0\\ 7 & -13\end{bmatrix}$$

$$\Rightarrow 13Y = \begin{bmatrix}39 & 26\\ -13 & 65\end{bmatrix}$$

$$\Rightarrow Y = \begin{bmatrix}3 & 2\\ -1 & 5\end{bmatrix}$$

Sol.16. (a)

$$A = \begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix} \quad Adj(A) = \begin{bmatrix}4 & -2\\ -1 & 3\end{bmatrix}$$

$$\therefore A\ Adj(A) = \begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}\begin{bmatrix}4 & -2\\ -1 & 3\end{bmatrix} = \begin{bmatrix}10 & 0\\ 0 & 10\end{bmatrix}$$

{Remember, A. Adj (A) = $|A|I_n$}

Sol.17. (c)

$$\because A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \text{ and } B = \begin{bmatrix}a & 0\\ 0 & b\end{bmatrix}$$

$$\therefore AB = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}a & 0\\ 0 & b\end{bmatrix} = \begin{bmatrix}a & 2b\\ 3a & 4b\end{bmatrix}$$

$$\text{And } BA = \begin{bmatrix}a & 0\\ 0 & b\end{bmatrix}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} = \begin{bmatrix}a & 2a\\ 3b & 4b\end{bmatrix}$$

If AB = BA, then

$$\begin{bmatrix}a & 2b\\ 3a & 4b\end{bmatrix} = \begin{bmatrix}a & 2a\\ 3b & 4b\end{bmatrix}$$

⇒a = b

From the above, it is clear that there exist infinitely many B's such that AB = BA.

Sol.18. (b)

$$\text{Let } A = \begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$$

$$\therefore |A| = -1 \text{ and adj (A)} = -\frac{1}{2}\begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix}$$

$$\text{Hence, } A^{-1} = \frac{1}{|A|}adj(A) = -\frac{1}{1}\begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix}$$

$$= \begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$$

Sol.19. (b)

Since, (AB)' = B' A' = BA

So, AB is not symmetric and

$(A^2 + B^2)' = (A')^2 + (B')^2 = A^2 + B^2$

Hence, $A^2 + B^2$ is symmetric

Sol.20. (d)

$\because |A| \neq 0$

So, A^{-1} exists

Now,

$$A^2 = \begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix}\begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = I$$

Sol.21. (c)

If AB = 0, then it may be concluded that either |A|=0 or |B| = 0,

Sol.22. (c)

Sol.23. (d)

Sol.24. (c)

$adj(A)^T = (adj\ A)^T$

Sol.25. (a)

We have $2A^3 + 2A^2 + 5A + I = 0$

$\Rightarrow I = -3A^3 - 2A^2 - 5A$

$\Rightarrow IA^{-1} = (-3A^2 - 2A^2 - 5A)A^{-1}$

$\Rightarrow A^{-1} = -3A^2 - 2A - 5I$

Sol.26. (c)

$$\text{By option, (c), let } B = \begin{bmatrix}5 & 2\\ 1 & 6\\ 4 & 3\end{bmatrix}$$

$$AB = \begin{bmatrix}5 & 6 & 1\\ 2 & -1 & 5\end{bmatrix}\begin{bmatrix}5 & 2\\ 1 & 6\\ 4 & 3\end{bmatrix} = \begin{bmatrix}35 & 49\\ 29 & 13\end{bmatrix}$$

Sol.27. (b)

If A'=A, then it is symmetric matrix

Sol.28. (d)

$$\therefore A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$$

$$\therefore A^{-1} = -\frac{1}{2}\begin{bmatrix}4 & -2\\ -3 & 1\end{bmatrix}$$

$$\Rightarrow [b_{ij}] = -\frac{1}{2}\begin{bmatrix}4 & -2\\ -3 & 1\end{bmatrix}$$

$$\Rightarrow b_{22} = -\frac{1}{2}$$

S.
OFFICIUM SUPREMUM

NDA PYQ

1. If the matrix $A = \begin{bmatrix} 2-x & 1 & 1 \\ 1 & 3-x & 0 \\ -1 & -3 & -x \end{bmatrix}$ is singular, then what is the solution set S?
(a) S = {0, 2, 3} (b) S = {–1, 2, 3}
(c) S = {1, 2, 3} (d) S = {2, 3}
[NDA (I) - 2011]

2. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ such that $A^2 = B$, then what is the value of α?
(a) –1 (b) 1
(c) 2 (d) 4
[NDA (I) - 2011]

3. If $A = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, then which of the following is/are correct?
I.AB is defined
II.BA is defined
III.AB = BA
Select the correct answer using the code given below:
(a) Only 1 (b) Only II
(c) I and II (d) I, II and III
[NDA (I) - 2011]

4. Consider the following statements.
I. The inverse of the square matrix, if it exits, is unique.
II. If A and B are singular matrices of order n, then AB is also a singular matrix of order n.
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2011]

5. If the matrix $A = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$ is such that $A^2 = I$, then which one of the following is correct?
(a) $\alpha=0, \beta=1$ or $\alpha=1, \beta=0$ (b) $\alpha=0, \beta\neq1$ or $\alpha\neq1, \beta=0$
(c) $\alpha=1, \beta\neq0$ or $\alpha\neq1, \beta=0$ (d) $\alpha \neq 0, \beta \neq 0$
[NDA (I) - 2011]

6. For what value of x does $(1\ 3\ 2)\begin{pmatrix} 1 & 3 & 0 \\ 3 & 0 & 2 \\ 2 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix} = (0)$ hold?
(a) –1 (b) 1
(c) 9/8 (d) –9/8
[NDA (II) - 2011]

7. Consider the following statements.
I. Every zero matrix is a square matrix
II.A matrix has a numerical value
III.A unit matrix is a diagonal matrix
Which of the above statement is/are correct?
(a) Only II (b) Only III
(c) Both II and III (d) Both I and III
[NDA (I) - 2012]

8. If a matrix A has inverse B and C, then which one of the following is correct?
(a) B may not be equal to C
(b) B should be equal to C
(c) B and C should be unit matrices
(d) Noun of the above
[NDA (I) - 2012]

9. If A is a square matrix such that $A^2 = I$· where I is the identity matrix, then what is the value of A^{-1}?
(a) A+ i (b) Null matrix
(c) A (d) Transpose of A
[NDA (I) - 2012]

10. What is the order of the product:
$[x\ y\ z]\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$?
(a) 3×1 (b) 1×1
(c) 1×3 (d) 3×3
[NDA (I) - 2012]

11. The sum and product of matrices A and B exist. Which of the following implications are necessarily true?
I. A and B are square matrices of same order
II. A and B are non-singular matrices.
Select the correct answer using the codes given below:
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2012]

12. If $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$, then what is the value of $B^{-1}A^{-1}$.
(a) $\begin{bmatrix} 1 & -3 \\ -1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$
(c) $\begin{bmatrix} -1 & 3 \\ -1 & -2 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & -3 \\ 1 & -2 \end{bmatrix}$
[NDA (I) - 2012]

13. A and B are two matrices such that AB=A and BA=B then what is B^2 equal to?
(a) B (b) A
(c) I (d) –I
Where I is the identity matrix
[NDA-2012(1)]

14. If $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 & 7 \\ 4 & 6 & 8 \end{bmatrix}$, then
Which one of the following is correct?
(a) B is the inverse of A (b) B is the adjoin of A
(c) B is the transpose of A (d) None of these
[NDA (II) - 2012]

15. The inverse of a diagonal matrix is a:
(a) Symmetric matrix (b) Skew-symmetric matrix
(c) Diagonal matrix (d) None of these
[NDA (II) - 2012]

16. If the matrix $\begin{bmatrix} \alpha & 2 & 2 \\ -3 & 0 & 4 \\ 1 & -1 & 1 \end{bmatrix}$ is not invertible, then:
(a) $\alpha = -5$ (b) $\alpha = 5$
(c) $\alpha = 0$ (d) $\alpha = 1$
[NDA (II) - 2012]

17. If the matrix AB is a zero matrix, then which one of the following is correct?

(a) A must be equal to zero matrix or B must be equal to zero matrix.
(b) A must be equal to zero matrix and B must be equal to zero matrix.
(c) It is not necessary that either A is zero matrixes or B is zero matrixes
(d) None of the above

[NDA (II) - 2012]

18. If the sum the matrices $\begin{bmatrix} x \\ x \\ y \end{bmatrix}, \begin{bmatrix} y \\ y \\ z \end{bmatrix}$ and $\begin{bmatrix} z \\ 0 \\ 0 \end{bmatrix}$ is the matrix $\begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix}$, then what is the value of y?

(a) –5 (b) 0
(c) 5 (d) 10

[NDA (II) - 2012]

19. A square matrix $[a_{ij}]$ such that $a_{ij} = 0$ for $i \neq j$ and $a_{ij} = k$ where k is a constant for i = j is called.
(a) Diagonal matrix but not scalar matrix
(b) Scalar matrix
(c) Unit matrix
(d) None of the above

[NDA (II) - 2012]

20. If A and B are two non-singular square matrices such that AB = A, then which one of the following is correct?
(a) B is an identity matrix
(b) $B = A^{-1}$
(c) $B = A^2$
(d) Determinant of B is zero

[NDA (I) - 2013]

21. If $\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$, then what is λ equal to ?
(a) 7 (b) –7
(c) 9 (d) –9

[NDA (II) - 2013]

22. If $A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ where $i = \sqrt{-1}$, then which one of the following is correct?
(a) AB = – C
(b) AB = C
(c) $A^2 = B^2 = C^2 = I$, where *I* is the identity matrix
(d) $BA \neq -C$

[NDA (II) - 2013]

23. Consider the following statements:
I. The product of two non-zero matrices can never be identity matrix.
II. The product of two non-zero matrices can never be zero matrix.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (II) - 2013]

24. Consider the following statements:
I. The matrix $\begin{pmatrix} 1 & 2 & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{pmatrix}$ is singular
II. The matrix $\begin{bmatrix} c & 2c & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{bmatrix}$, in non-singular
Which of the above statement is/are correct.
(a) Only I (b) Only II
(c) Both I and II (d) Either I or II

[NDA (II) - 2013]

25. If $2A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, then what is A^{-1} equal to?
(a) $\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$ (b) $\frac{1}{2}\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$
(c) $\frac{1}{4}\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$ (d) None of these

[NDA (II) - 2013]

26. If A is a square matrix of order 3 with $|A| \neq O$, then which one of the following is correct?
(a) $|\text{adj } A| = |A|$ (b) $|\text{adj } A| = |A|^2$
(c) $|\text{adj } A| = |A|^3$ (d) $|\text{adj } A|^2 = |A|$

[NDA(II)-2013]

27. Consider the following statements in respect of the matrix A $= \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix}$:
I. The matrix A is skew symmetric
II. The matrix A is symmetric
III The matrix A is invertible
Which of the above statements is /are correct?
(a) Only I (b) Only III
(c) I and II (d) II and III

[NDA (I) - 2014]

28. Consider two matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}$ which one of the following is correct?
(a) B is the right inverse of A
(b) B is the left inverse of A
(c) B is the both sided inverse of A
(d) None of the above

[NDA (I) - 2014]

29. If A is any matrix, then the product AA is defined only when A is a matrix of order $m \times n$, where:
(a) $m > n$ (b) $m < n$
(c) $m = n$ (d) $m \leq n$

[NDA (I) - 2014]

30. If, A and B be two matrices such that AB = A and BA = B. Then, which of the following statements are correct?
1. $A^2 = A$
2. $B^2 = B$
3. $(AB)^2 = AB$
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

[NDA (II) - 2014]

31. If the matrix A is such that $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$, then what is a equal to?
(a) $\begin{pmatrix} 1 & 4 \\ 0 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$
(c) $\begin{pmatrix} -1 & 4 \\ 0 & -1 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & -4 \\ 0 & -1 \end{pmatrix}$

[NDA (II) - 2014]

32. From the matrix equation AB = AC, where A, B and C are the square matrices of same order, we can conclude B = C provided:
(a) A is non-singular
(b) A is singular
(c) A is symmetric
(d) A is skew symmetric
[NDA (II) - 2014]

33. If A = $\begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is symmetric, then what is x equal to?
(a) 2 (b) 3
(c) –1 (d) 5
[NDA (II) - 2014]

34. The matrix $\begin{bmatrix} 0 & -4+i \\ 4+i & 0 \end{bmatrix}$ is:
(a) Symmetric (b) Skew-symmetric
(c) Hermitian (d) Skew-hermitian
[NDA (I) - 2015]

35. Which one of the following matrices is an elementary matrix?
(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 2 \end{bmatrix}$
[NDA (I) - 2015]

36. If A = $\begin{bmatrix} 2 & 7 \\ 1 & 5 \end{bmatrix}$, then what is $A + 3A^{-1}$ equal to?
(a) 3I (b) 5I
(c) 7I (d) none of these
[NDA (I) - 2015]

37. If X = $\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, B = $\begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix}$ and A = $\begin{bmatrix} p & q \\ r & s \end{bmatrix}$ satisfy the equation AX = B, then the matrix A is equal to:
(a) $\begin{bmatrix} -7 & 26 \\ 1 & -5 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & 26 \\ 4 & 17 \end{bmatrix}$
(c) $\begin{bmatrix} -7 & -4 \\ 26 & 13 \end{bmatrix}$ (d) $\begin{bmatrix} -7 & 26 \\ -6 & 23 \end{bmatrix}$
[NDA (I) - 2015]

38. Let A= $\begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}$, B = $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and C = $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$. If AB = C, then what is A^2 equal to?
(a) $\begin{bmatrix} 6 & -10 \\ 4 & 26 \end{bmatrix}$ (b) $\begin{bmatrix} -10 & 5 \\ 4 & 24 \end{bmatrix}$
(c) $\begin{bmatrix} -5 & -6 \\ -4 & -20 \end{bmatrix}$ (d) $\begin{bmatrix} -5 & -7 \\ -5 & 20 \end{bmatrix}$
[NDA (I) - 2015]

39. If E(θ)= $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ then E(α) E(β) is equal to:
(a) E(αβ) (b) E(α–β)
(c) E(α+β) (d) –E (α+β)
[NDA (I) - 2015]

40. If A = $\begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 4 \end{bmatrix}$, then the matrix X for which 2x + 3A = 0 holds true is:
(a) $\begin{bmatrix} -\frac{3}{2} & 0 & -3 \\ -3 & -\frac{9}{2} & -6 \end{bmatrix}$ (b) $\begin{bmatrix} -\frac{3}{2} & 0 & -3 \\ 3 & -\frac{9}{2} & -6 \end{bmatrix}$
(c) $\begin{bmatrix} \frac{3}{2} & 0 & 3 \\ 3 & -\frac{9}{2} & 6 \end{bmatrix}$ (d) $\begin{bmatrix} -\frac{3}{2} & 0 & 3 \\ -3 & \frac{9}{2} & -6 \end{bmatrix}$
[NDA (II) - 2015]

41. The matrix A = $\begin{bmatrix} 1 & 3 & 2 \\ 1 & x-1 & 1 \\ 2 & 7 & x-3 \end{bmatrix}$ will have inverse for every real number x except for:
(a) x= $\frac{11\pm\sqrt{5}}{2}$ (b) x= $\frac{9\pm\sqrt{5}}{2}$
(c) x= $\frac{11\pm\sqrt{3}}{2}$ (d) x= $\frac{9\pm\sqrt{3}}{2}$
[NDA (II) - 2015]

42. If A = $\begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and B = $\begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ then the following is/are correct?
1. A and B commute
2. AB is a null matrix
Select the correct answer using the codes given below:
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (II) - 2015]

43. Consider the following in respect of the matrix A = $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$:
1. $A^2 = -A$
2. $A^3 = 4A$
Which of the above is /are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (I) - 2016]

44. If A is a square matrix, then what is adj (A^{-1}) – $(\text{adj}A)^{-1}$ equal to?
(a) 2|A| (b) Null matrix
(c) Unit matrix (d) None of these
[NDA (I) - 2016]

45. What is [x y z] $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ equal to?
(a) [ax + hy + gz h + b + f g + f + c]
(b) $\begin{bmatrix} a & h & g \\ hx & by & fz \\ g & f & c \end{bmatrix}$

(c) $\begin{bmatrix} ax & hy & gz \\ hx & by & fz \\ gx & fy & cz \end{bmatrix}$

(d) [ax + hy + gz hx + by + fz gx + fy + cz]

[NDA (II) - 2016]

46. If $m = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $n = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, then what is the value of the determinant of $m \cos\theta - n \sin\theta$?

(a) –1 (b) 0
(c) 1 (d) 2

[NDA (II) - 2016]

47. If $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then which of the following are correct?

1. $f(\theta) \times f(\phi) = f(\theta+\phi)$.
2. The value of the determinant of the matrix $f(\theta) \times f(\phi)$ is 1.
3. The determinant of f(x) is an even number.

Select the correct answer using the codes given below:

(a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2, and 3

[NDA (II) - 2016]

48. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$, then which of the following is/are correct?

1. $AB(A^{-1}B^{-1})$ is a unit matrix.
2. $(AB)^{-1} = A^{-1}B^{-1}$

Select the correct answer using the codes given below:

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2016]

49. If $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, then what is AA^T equal to (where A^T is the transpose of A)?

(a) Null matrix (b) Identity matrix
(c) A (d) –A

[NDA (I) - 2017]

50. $A = \begin{bmatrix} x+y & y \\ x & x-y \end{bmatrix}$, $B = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and $C = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$. If AB = C, then what is A^2 equal to?

(a) $\begin{bmatrix} 4 & 8 \\ -4 & -16 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & -4 \\ 8 & -16 \end{bmatrix}$

(c) $\begin{bmatrix} -4 & -8 \\ 4 & 12 \end{bmatrix}$ (d) $\begin{bmatrix} -4 & -8 \\ 8 & 12 \end{bmatrix}$

[NDA (I) - 2017]

51. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then what is A^3 equal to?

(a) $\begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$ (b) $\begin{bmatrix} \cos^3\theta & \sin^3\theta \\ -\sin^3\theta & \cos^3\theta \end{bmatrix}$

(c) $\begin{bmatrix} \cos 3\theta & -\sin 3\theta \\ -\sin 3\theta & \cos\theta \end{bmatrix}$ (d) $\begin{bmatrix} \cos^3\theta & \sin^3\theta \\ -\sin^3\theta & \cos^3\theta \end{bmatrix}$

[NDA (I) - 2017]

52. What is the order of :

$[x\ y\ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$?

(a) 3×1 (b) 1 × 1
(c) 1 × 3 (d) 3 × 3

[NDA (I) - 2017]

53. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the value of A^4 is:

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

[NDA (I) - 2017]

54. If $A = \begin{bmatrix} 4i-6 & 10i \\ 14i & 6+4i \end{bmatrix}$ and $k = \frac{1}{2i}$, where $i = \sqrt{-1}$ then kA is equal to:

(a) $\begin{bmatrix} 2+3i & 5 \\ 7 & 2-3i \end{bmatrix}$ (b) $\begin{bmatrix} 2-3i & 5 \\ 7 & 5+3i \end{bmatrix}$

(c) $\begin{bmatrix} 2-3i & 7 \\ 5 & 3+3i \end{bmatrix}$ (d) $\begin{bmatrix} 2+3i & 5 \\ 7 & 2+3i \end{bmatrix}$

[NDA (II) - 2017]

55. If A is a square matrix, then the value of adj A^T – $(\text{adj } A)^T$ is equal to:

(a) A
(b) 2|A|*l*, where *l* is the indentify matrix
(c) Null matrix whose order is same as that of A
(d) Unit matrix whose order is same as that of A

[NDA (II) - 2017]

56. If a, b, c are non-zero real numbers, then the inverse of the matrix, $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ is equal to:

(a) $\begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$ (b) $\frac{1}{abc}\begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$

(c) $\frac{1}{abc}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (d) $\frac{1}{abc}\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

[NDA (II) - 2017]

57. The ad joint of the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$ is:

(a) $\begin{bmatrix} -1 & 6 & 2 \\ -2 & 1 & -4 \\ 6 & 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 6 & -2 \\ -2 & 1 & 4 \\ 6 & -3 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 6 & 1 & 2 \\ 4 & -1 & 2 \\ 6 & 3 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} -6 & 2 & 1 \\ 4 & -2 & 1 \\ 3 & 1 & -6 \end{bmatrix}$

[NDA (II) - 2017]

58. If $A = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$, then which one of the following is correct?

(a) $A^2 = -2A$ (b) $A^2 = -4A$
(c) $A^2 = -3A$ (d) $A^2 = 4A$

[NDA (II) - 2017]

59. The matrix A has x rows and x + 5 columns. The matrix B has y rows and 11−y columns both AB and BA exist. What are the values of x and y respectively?
(a) 8 and 3 (b) 3 and 4
(c) 3 and 8 (d) 8 and 8

[NDA (II) - 2017]

60. If α and β are the roots of the equation $1 + x + x^2 = 0$, then the matrix product $\begin{bmatrix} 1 & \beta \\ \alpha & \alpha \end{bmatrix}\begin{bmatrix} \alpha & \beta \\ 1 & \beta \end{bmatrix}$ is equal to:

(a) $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix}$

[NDA (II) - 2017]

61. If $A=\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ and $A^2 - kA - I_2 = O$, where I_2 is the 2×2 identity matrix, then what is the value of k?
(a) 4 (b) −4
(c) 8 (d) −8

[NDA (I) - 2018]

62. If A is a 2×3 matrix and AB is a 2×5 matrix, then B must be a
(a) 3×5 matrix (b) 3×3 matrix
(c) 3×2 matrix (d) 5×2 matrix

[NDA (I) - 2018]

63. What is the inverse of the matrix

$A = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(a) $\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (b $\begin{pmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$ (d) $\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

[NDA (I) - 2018]

64. Consider the following in respect of matrices A and B of same order:
I.$A^2 - B^2 = (A+B)(A-B)$
II.$(A-I)(I+A) = 0 \Leftrightarrow A^2 = I$
Where I is the identity matrix and O is the null matrix.
Which of the above is/are correct?
(a) I only (b) 2 only
(c) Both I and II (d) Neither 1 nor II

[NDA (II) - 2018]

65. Let matrix B be the adjoin of a square matrix A, ℓ be the identity matrix of same order as A. If k ($\neq$ 0) is the determinant of the matrix A, then what is AB equal to?
(a) ℓ (b) $k\ell$
(c) $k^2\ell$ (d) $(1/k)\ell$

[NDA (II) - 2018]

66. Consider the following in respect of matrices A, B and C of same order:
I.(A+B+C)’ = A’+B’+C’
II.(AB)’ = A’B’
III.(ABC)’ = C’B’A’
Where A’ is the transpose of the matrix A. Which of the above are correct?
(a) I and II only (b) II and III only
(c) I and III only (d) I, II and III only

[NDA (II) - 2018]

67. What should be the value of x so that the matrix $\begin{pmatrix} 2 & 4 \\ -8 & x \end{pmatrix}$ does not have an inverse?
(a) 16 (b) −16
(c) 8 (d) −8

[NDA (II) - 2018]

68. If A and B are two invertible square matrices of same order, then what is $(AB)^{-1}$ equal to?
(a) $B^{-1}A^{-1}$ (b) $A^{-1}B^{-1}$
(c) $B^{-1}A$ (d) $A^{-1}B$

[NDA (II) - 2018]

69. What is the ad joint of the matrix:
$\begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ -\sin(-\theta) & \cos(-\theta) \end{pmatrix}$

(a) $\begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ (b) $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

(c) $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ (d) $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

[NDA (II) - 2018]

70. A square matrix A is called orthogonal if.
(a) $A=A^2$ (b) $A’=A^{-1}$
(c) $A= A^{-1}$ (d) A=A’

[NDA (II) - 2018]

71. If A is an identity matrix of order 3, then its inverse (A^{-1}).
(a) is equal to null matrix
(b) is equal to A
(c) is equal to 3A
(d) does not exist

[NDA-2019(1)]

72. If $B = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 4 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ than what is adjoint of B equal to?

(a) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & -1 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 8 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (d) Does not exist

[NDA (I) - 2019]

73. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ then the matrix A is
(a) Singular matrix (b) Involutory matrix
(c) Nilpotent matrix (d) Idempotent matrix

[NDA (I) - 2019]

74. If $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ then the expression $A^3 - 2A^2$ is

(a) a null matrix (b) an identity matrix
(c) equal to A (d) equal to – A

[NDA (II) - 2019]

75. If $A=\begin{bmatrix}1&2\\2&3\\3&4\end{bmatrix}$ and $B=\begin{bmatrix}1&2\\2&1\end{bmatrix}$, then

Which one of the following is correct?
(a) Both AB and BA exists
(b) Neither AB nor BA exists
(c) AB exists but BA not exists
(d) AB does not exists but BA exists

[NDA (II) - 2019]

76. Let $A=\begin{bmatrix}a&h&g\\h&b&f\\g&f&c\end{bmatrix}$, $B=\begin{bmatrix}x\\y\\z\end{bmatrix}$ then what is AB equal to?

(a) $\begin{bmatrix}ax+hy+gz\\y\\z\end{bmatrix}$

(b) $\begin{bmatrix}ax+hy+gz\\hx+by+fz\\z\end{bmatrix}$

(c) $\begin{bmatrix}ax+hy+gz\\hx+by+fz\\gx+fy+cz\end{bmatrix}$

(d) $[ax + hy + gz \quad hx + by + fz \quad gx + fy + cz]$

[NDA 2020]

77. If A is a matrix of order 3 × 5 and B is a matrix of order 5 × 3, then the order of AB and BA will respectively be
(a) 3 × 3 and 3 × 3 (b) 3 × 5 and 5 × 3
(c) 3 × 3 and 5 × 5 (d) 5 × 3 and 3 × 5

[NDA 2020]

78. If matrix $A=\begin{bmatrix}1-i&i\\-i&1-i\end{bmatrix}$ then which one the following is correct?
(a) A is hermition
(b) A is skew hermition
(c) $(\overline{A})^T+A$ is hermition
(d) $(\overline{A})^T+A$ is skew hermition

[NDA 2020]

79. For how many values of k, is the matrix $\begin{bmatrix}0&k&4\\-k&0&-5\\-k&k&-1\end{bmatrix}$ is singular?
(a) Only one (b) Only two
(c) Only 4 (d) Infinite

[NDA 2020]

80. Let $A=\begin{bmatrix}x+y&y\\2x&x-y\end{bmatrix}$, $B=\begin{bmatrix}2\\-1\end{bmatrix}$ and $C=\begin{bmatrix}3\\2\end{bmatrix}$ If AB=C, then what is the value of determinant of A?
(a) −10 (b) −14
(c) −24 (d) −34

[NDA 2020]

81. Consider the following in respect of a non singular matrix of order 3
I. A(adjA) = (adjA)A
II. |adj A| = |A|
Where I is the identity matrix and O is the null matrix.
Which of the above is/are correct?
(a) I only (b) 2 only
(c) Both I and II (d) Neither 1 nor II

[NDA 2020]

82. If A and B are two matrices such that AB of order n × n, then which one of the following is correct?
(a)A and B should be square matrices of same order
(b) Either A or B should be a square matrix.
(c) Both A and B should be same order
(d) Order of A and B need not be same.

[NDA (I) 2021]

83. How many matrices of different orders are possible with elements comprising all prime numbers less that 30?
(a) 2 (b) 3
(c) 4 (d) 6

[NDA (I) 2021]

84. If $A=\begin{vmatrix}1&a\\0&1\end{vmatrix}$ where A∈N, then what is $A^{100}-A^{50}-2A^{25}$ equal to?
(a) –2I (b) –I
(c) 2I (d) I
Where I is the identify matrix.

[NDA (II) 2021]

85. How many distinct matrices exist with all four entries taken from {1,2} ?
(a) 16 (b) 24
(c) 32 (d) 48

[NDA (II) 2021]

86. Consider the following in respect of the matrix:
$A=\begin{vmatrix}1&1&1\\1&1&1\\1&1&1\end{vmatrix}$
1.Inverse of A does not exist
2. $A^3 = A$
3. $3A = A^2$
Which of the above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (II) 2021]

87. The inverse of a matrix A is given by $\begin{bmatrix}-2&1\\\frac{3}{2}&-\frac{1}{2}\end{bmatrix}$. What is A equal to?
(a) $\begin{bmatrix}1&2\\3&4\end{bmatrix}$ (b) $\begin{bmatrix}1&-2\\-3&4\end{bmatrix}$
(c) $\begin{bmatrix}1&2\\3&-4\end{bmatrix}$ (d) $\begin{bmatrix}-1&2\\3&4\end{bmatrix}$

[NDA (II) 2021]

88. Let $A=\begin{bmatrix}0&2\\-2&0\end{bmatrix}$ and $(mI + nA)^2 = A$ where m, n are positive real numbers and I is the identity matrix. What is (m + n) equal to:
(a) 0 (b) $\frac{1}{2}$

(c) 1 (d) $\frac{3}{2}$

[NDA (II) 2021]

89. Consider the following in respect of the matrices:

A=[m n], B=[–n –m] and C = $\begin{bmatrix} m \\ -m \end{bmatrix}$

1.CA = CB
2.AC = BC
3.C(A + B) = CA + CB

Which of the above statement is/are correct?

(a)1 only (b) 2 only
(c) 2 and 3 (d) 1 and 2

[NDA (I) - 2022]

90. If A = $\begin{bmatrix} 2\sin\theta & \cos\theta & 0 \\ -2\cos\theta & \sin\theta & 0 \\ -1 & 1 & 1 \end{bmatrix}$, then what is A (adj. A) equal to?

(a) Null matrix (b) –I
(c) I (d) 2I

Where I is the identity matrix.

[NDA (I) - 2022]

91. For what value of k is the matrix $\begin{bmatrix} 2\cos 2\theta & 2\cos 2\theta & 6 \\ 1-2\sin^2\theta & 2\cos^2\theta-1 & 3 \\ k & 2k & 1 \end{bmatrix}$ singular ?

(a) 0 only (b) 1 only
(c) 2 only (d) Any real value

[NDA (I) - 2022]

92. Let A be a non-singular matrix and B = adjA. Which of the following statement is/are correct?

1.AB = BA
2.AB is a scalar matrix
3.AB can be null matrix

Select the correct answer using the code gien below:

(a) 1 only (b) 1 and 2 only
(c) 2 only (d) 1, 2 and 3

[NDA (I) - 2022]

93. Consider the following statements in respect of square matrices A and B of same order

1.If AB is a null matrix, then at least one of A and B is a null matrix
2.If AB is an identity matrix, then BA = AB

Which of the above statement is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) neither 1 nor 2

[NDA (I) - 2022]

94. If A is the identity matrix of order 3 and B is its transpose, then what is the value of determinant of the matrix C = A+ B?

(a) 1 (b) 2
(c) 4 (d) 8

[NDA (I) - 2022]

95. Let A and B be non-singular matrices of the same order such that AB =A and BA = B. Which of the following statements is/are correct?

1.A^2=A
2.AB^2=A^2B

Select the correct answer using the code given below:

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

Consider the following for the next three (03) items that follow:

Let A = $\begin{pmatrix} 0 & \sin^2\theta & \cos^2\theta \\ \cos^2\theta & 0 & \sin^2\theta \\ \sin^2\theta & \cos^2\theta & 0 \end{pmatrix}$ and A = P + Q where P is symmetric matrix and Q is skew-symmetric matrix.

96. What is P equal to?

(a) $\begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

(c) $\cos 2\theta \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$ (d) $\cos 2\theta \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/2 & 1/2 & 0 \end{pmatrix}$

[NDA 2022 (II)]

97. What is Q equal to?

(a) $\begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

(c) $\cos 2\theta \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$ (d) $\cos 2\theta \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/2 & 1/2 & 0 \end{pmatrix}$

[NDA 2022 (II)]

98. What is the minimum value of determinant of A?

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{3}{4}$ (d) 1

[NDA 2022 (II)]

99. If A = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & \sin\theta & -\cos\theta \end{pmatrix}$, then which of the following are correct?

1. A + adjA is a null matrix
2. A^{-1} + adjA is null matrix
3. A – A^{-1} is a null matrix

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA 2022 (II)]

100. If X is a matrix of order 3 × 3, Y is a matrix of order 2 × 3 and Z is a matrix of order 3 × 2, then which of the following are correct?

1.(ZY)X is a square matrix having 9 entries
2.Y(XZ) is a square matrix having 4 entries
3.X(YZ) is not defined

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA 2022 (II)]

101. Let a and B be symmetric matrices of same order, then which one of the following is correct regarding (AB – BA)?

1.Its diagonal entries are equal but nonzero

2.The sum of its non-diagonal entries is zero
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2
[NDA – 2023 (1)]

102. Consider the following statements in respect of square matrices A, B, C each of same order n:
1.AB = AC $\Rightarrow$ B = C if A is non singular
2.If BX = CX for every column matrix X having n rows then B = C.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2
[NDA – 2023 (1)]

103. Let AX = B be a system of 3 linear equations with 3-unknowns. Let X_1 and X_2 be its two distinct solutions. If the combination $aX_1 + bX_2$ is a solution of AX = B; where a, b are real numbers, then which one of the following is correct?
(a) a = b (b) a + b = 1
(c) a + b = 0 (d) a – b = 1
[NDA – 2023 (1)]

104. If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then what is the value of det (I+AA'), where I is the 3 × 3 identity matrix?
(a) 15 (b) 6
(c) 0 (d) –1
[NDA-2023 (2)]

105. If $A = \begin{bmatrix} 0 & 3 & 4 \\ -3 & 0 & 5 \\ -4 & -5 & 0 \end{bmatrix}$, then which one of the following statements is correct?
(a) A^2 is symmetric matrix with $\det(A^2)=0$
(b) A^2 is symmetric matrix with det $(A^2) \neq 0$.
(c) A^2 is skew-symmetric matrix with $\det(A^2)=0$
(d) A^2 is skew-symmetric matrix with $\det(A^2)\neq 0$
[NDA-2023 (2)]

106. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, then which one of the following statements are correct?
1.A^n will always be singular for any positive integer n
2.A^n will always be a diagonal matrix for any positive integer n
3.A^n will always be a symmetric matrix for any positive integer n.
Select the correct answer using the code given below:
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
[NDA-2023 (2)]

107. If $A = \begin{vmatrix} 2 & -1 & 0 \\ -1 & 3 & 0 \\ 1 & 0 & 1 \end{vmatrix}$, then what is the value of det[adj(adjA)]?
(a) 5 (b) 25
(c) 125 (d) 625
[NDA-2023 (2)]

108. If $A = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$, then what is $23A^2 - 19A^2 - 4A$ equal to?
(a) null matrix of order 3 (b) identity matrix of order 3
(c) $\begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix}$ (d) $\begin{vmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{vmatrix}$
[NDA-2023 (2)]

109. What is the number of different matrices, each having 4 entries that can be formed using 1, 2, 4, 5, (repletion is allowed)?
(a) 72 (b) 216
(c) 254 (d) 768
[NDA-2024 (1)]

110. If $A = \begin{bmatrix} \sin 2\theta & 2\sin^2\theta - 1 & 0 \\ \cos 2\theta & 2\sin\theta\cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then which of the following statements is/are correct?
1. $A^{-1} = \text{adj}A$
2. A is skew-symmetric matrix
3. $A^{-1} = A^T$
Select the correct answer using the code given below:
(a) 1 only (b) 1 and 2
(c) 1 and 3 (d) 2 and 3
[NDA-2024 (1)]

111. Consider the following statements in respect of two non-singular matrices A and B are of same order n :
1. adj(AB) = (adjA)(adjB)
2. adj(AB) = adj(BA)
3. (AB)adj(AB) – $|AB|I_n$ is a null matrix of order n
How many of the above statements are correct?
(a) None (b) Only one statement
(c) Only two statements (d) All three statements
[NDA-2024 (1)]

112. Consider the following statements in respect of a non-singuar matrix A of order n :
1. $A(\text{adj}A^T) = A(\text{adj}A)^T$
2. If $A^2 = A$, then A is identity matrix of order n
3. If $A^3 = A$, then A is identity matrix of order n
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
[NDA-2024 (1)]

113. Consider the following statements in respect of a skew-symmetric matrix A of order 3:
1. All diagonal elements are zero
2. The sum of all the diagonal elements of the matrix is zero
3. A is orthogonal matrix
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below

Let $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

114. What is A(adjA) equal to ?
(a) $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/2 \end{bmatrix}$ [redacted]

[NDA-2024 (2)]

115. What is A^{-1} equal to ?

(a) $\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$ (b) $\begin{bmatrix} 1/2 & -1/2 & 0 \\ -1 & 3/2 & -2 \\ -1 & 3/2 & -3/2 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & -2 & 0 \\ -4 & 6 & -8 \\ -4 & 6 & -6 \end{bmatrix}$ (d) $\begin{bmatrix} 1/5 & -1/5 & 0 \\ -2/5 & 3/5 & -4/5 \\ -2/5 & 3/5 & -3/5 \end{bmatrix}$

[NDA-2024 (2)]

116. Consider the following in respect of matrices

$$P = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \text{ and } Q = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

I. PQ is null matrix.
II. QP is an identity matrix of order 3
III. PQ = QP
Which of the above is/are correct?
(a) I only (b) II only
(c) I and III (d) II and III

[NDA-2024 (2)]

117. Let X be a matrix of order 3 × 3, Y be a matrix of order 2 × 3 and Z be a matrix of order 3 × 2. Which of the following statements are correct?
I. (ZY)X is defined and is a square matrix of order 3
II. Y(XZ) is defined and is a square matrix of order 2
III. X(YZ) is not defined.
Select the answer using the code given below
(a) I and II only (b) II and III only
(c) I and III only (d) I, II and III

[NDA-2024 (2)]

118. Let A and B be two square matrices of same order. If AB is a null matrix, then which one of the following is correct?
(a) Both A and B are null matrices
(b) Either A or B is a null matrix
(c) B is a null matrix if A is a non-singular matrix
(d) Both A and B are singular matrices

[NDA-2024 (2)]

119. If

$$\begin{bmatrix} x & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = \begin{bmatrix} 45 \end{bmatrix}$$

Then which one of the following is a value of x?
(a) – 2 (b) – 1
(c) 0 (d) 1

[NDA-2025 (1)]

120. If

$$A = \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix}$$

where x, y, z are integers, is an orthogonal matrix, then what is the value of $x^2 + y^2 + z^2$?
(a) 0 (b) 1
(c) 2 (d) 3

[NDA-2025 (1)]

121. Consider the following in respect of a non-singular matrix M:
I. $|M^2| = |M|^2$
II. $|M| = |M^{-1}|$
III. $|M| = |M^T|$
How many of the above are correct?
(a) None (b) One
(c) Two (d) All three

[NDA-2025 (1)]

122. If $f(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

Then, what is $\{f(\pi)\}^2$ equal to?

(a) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

[NDA-2025 (1)]

123. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then what is $A^2 - 4A$ is equal to?

(a) $-5I_3$ (b) $-I_3$
(c) I_3 (d) $5I_3$
Where, I_3 is the identity matrix of order 3

[NDA-2025 (1)]

124. If $A = \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix}$

Where x, y, z are integers, is an orthogonal matrix, then what is A^2 equal to?
(a) Null matrix (b) Identity matrix
(c) A (d) –A

[NDA-2025 (1)]

125. Consider the following in respect of non-singular matrices A and B:
I. $(AB)^{-1} = A^{-1}B^{-1}$
II. $(BA)(AB)^{-1} = I$, where I is the identity matrix
III. $(AB)^T = A^T B^T$
How many of the above are correct?
(a) None (b) One
(c) Two (d) All three

[NDA-2025 (2)]

ANSWER KEY

1.	a	**2.**	b	**3.**	d	**4.**	c	**5.**	a	**6.**	d	**7.**	b	**8.**	b	**9.**	c	**10.**	b
11.	a	**12.**	b	**13.**	a	**14.**	c	**15.**	c	**16.**	a	**17.**	c	**18.**	b	**19.**	b	**20.**	a
21.	b	**22**	a	**23.**	d	**24.**	a	**25.**	d	**26.**	b	**27.**	a	**28.**	b	**29.**	c	**30.**	d
31.	a	**32.**	a	**33.**	d	**34.**	d	**35.**	b	**36.**	c	**37.**	a	**38.**	a	**39.**	c	**40.**	d
41.	a	**42.**	b	**43.**	b	**44.**	b	**45.**	d	**46.**	c	**47.**	a	**48.**	d	**49.**	b	**50.**	d
51.	a	**52.**	b	**53.**	a	**54.**	a	**55.**	c	**56.**	a	**57.**	b	**58.**	b	**59.**	c	**60.**	b
61.	a	**62.**	a	**63.**	a	**64.**	b	**65.**	b	**66.**	c	**67.**	b	**68.**	a	**69.**	a	**70.**	b
71.	b	**72.**	a	**73.**	b	**74.**	a	**75.**	c	**76.**	c	**77.**	c	**78.**	c	**79.**	d	**80.**	b
81.	a	**82.**	d	**83.**	c	**84.**	a	**85.**	d	**86.**	c	**87.**	a	**88.**	d	**89.**	c	**90.**	d
91.	d	**92.**	b	**93.**	b	**94.**	d	**95.**	c	**96.**	a	**97.**	d	**98.**	a	**99.**	d	**100.**	d
101.	b	**102.**	c	**103.**	b	**104.**	a	**105.**	a	**106.**	b	**107.**	d	**108.**	a	**109.**	d	**110.**	c
111.	b	**112.**	a	**113.**	a	**114.**	d	**115.**	a	**116.**	c	**117.**	d	**118.**	c	**119.**	d	**120.**	b
121.	c	**122.**	d	**123.**	d	**124.**	b	**125.**	a										

Solutions

Sol.1. (a)
For the singular matrix,
$\begin{vmatrix} 2-x & 1 & 1 \\ 1 & 3-x & 0 \\ -1 & -3 & -x \end{vmatrix}$ =0 (expand with respect to R_1)
$\Rightarrow(2-x)\,[x(x-3)]-[-x]+[-3+(3-x)]=0$
$\Rightarrow x(x-3)\,(x-2)=0$
$\Rightarrow x=0,2,3$
So, the solution set is, S = {0, 2, 3}

Sol.2. (b)
$A=\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}, B=\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$
$A^2=\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}=\begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}$
Given, $A^2 = B$
$\begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}=\begin{bmatrix} 4 & -2 \\ 2 & 1 \end{bmatrix}$
On comparing, $\alpha+1=2$
$\Rightarrow \alpha=1$

Sol.3. (d)
Given matrices are,
$A=\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}_{2\times2}, B=\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}_{2\times2}$
Since, the order of both matrices are same
So, AB and BA is well defined
Now, AB =
$\begin{bmatrix} 3\times1+1\times0 & 3\times1+1\times2 \\ 0\times1+4\times0 & 0\times1+4\times2 \end{bmatrix}=\begin{bmatrix} 3 & 5 \\ 0 & 8 \end{bmatrix}$
and $AB=\begin{bmatrix} 1\times3+1\times0 & 1\times1+4\times1 \\ 3\times0+2\times0 & 0\times1+2\times4 \end{bmatrix}=\begin{bmatrix} 3 & 5 \\ 0 & 8 \end{bmatrix}$
$\Rightarrow$AB= BA

Sol.4. (c)
I.If the inverse of a square matrix exists, then it will be unique
II.If two matrices A and B are singular of order n, then the product of both matrix is also a singular matrix of same order.

Sol.5. (a)
$A^2=\begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}\begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}=I$
$\Rightarrow\begin{bmatrix} \alpha^2+\beta^2 & 2\alpha\beta \\ 2\alpha\beta & \alpha^2+\beta^2 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
$\Rightarrow\alpha^2+\beta^2=1,\ 2\alpha\beta=0$
$\Rightarrow\alpha=0,\beta=1$ or $\alpha=1,\beta=0$

Sol.6. (d)
Given, $(1,2,3)_{1\times3}\begin{pmatrix} 1 & 3 & 0 \\ 3 & 0 & 2 \\ 2 & 0 & 1 \end{pmatrix}_{3\times3}\begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix}_{3\times1}=(0)_{1\times1}$
$\Rightarrow(1+9+4\quad 3+0+0\quad 0+6+2)_{1\times3}\begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix}_{3\times1}=(0)_{1\times1}$
$\Rightarrow(14\ 3\ 8)_{1\times3}\begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix}_{3\times1}=(0)_{1\times1}$
$\Rightarrow(0+9+8x)=(0)$
$\Rightarrow(8x+9)=(0)$
On comparing,
$8x+9=0\Rightarrow x=-\frac{9}{8}$

Sol.7. (b)
I. Every zero-matrix is not necessarily square matrix
II. A matrix does not have a numerical value while every determinant have a numerical value
III. Unit matrix is a diagonal matrix and scalar matrix also.
Only statement III is correct.

Sol.8. (b)
We know that, every matrix possesses a unique inverse. Hence, B and C should be equal.

Sol.9. (c)
Given condition is $A^2 = I$
$\Rightarrow A^{-1}.A^2=A^{-1}.\,I$
$\Rightarrow A^{-1}\,(A.A)=A^{-1}\qquad(\therefore A.I=a)$
$\Rightarrow(A^{-1}A).A=A^{-1}$
$\Rightarrow I.A=A^{-1}\qquad(\therefore A^{-1}A=I)$
$\Rightarrow A^{-1}=A$

Sol.10. (b)
Here,
$[x\ y\ z]_{1\times3}\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}_{3\times3}\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3\times1}$
Order of matrix = 1 × 3 : 3 × 3 : 3 × 1 = 1 × 1

Sol.11. (a)
The sum and product of matrices A and B exist, if A and B are square matrices of same order
It is not necessary that A and B are non-singular matrices for addition and product of two matrices.

Sol.12. (b)
Given, $A=\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ and $B=\begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$
Here, $|A|=\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}=1-2=-1$
adj. (A) $\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}^T=\begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}$
$A^{-1}=\frac{adj.A}{|A|}=-1\begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}$
$A^{-1}=\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$
Here, $|B|=\begin{vmatrix} 0 & -1 \\ 1 & -2 \end{vmatrix}=0-(-1)=1$
adj (B) = $\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}^T=\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$
$B^{-1}=\frac{adj.B}{|B|}=1,\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}; B^{-1}=\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$
$\therefore B^{-1}A^{-1}=$
$\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}=\begin{bmatrix} -1+1 & 4-1 \\ 1+0 & -2+0 \end{bmatrix}$
$=\begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$

Sol.13. (a)
AB = A
Multiply with B at left side
BAB = BA
(BA)B = BA
$B^2 = B$

Sol.14. (c)
B is the transpose of A

Sol.15. (c)
We know that by the property of diagonal matrix
At A = diagonal (a_1, a_2, a_3)
Then, A^{-1} = Inverse of A
=Diagonal $\left(a_1^{-1}, a_2^{-1}, a_3^{-1}\right)$
=Diagonal$\left(\frac{1}{a_1},\frac{1}{a_2},\frac{1}{a_3}\right)$
Hence, the inverse of diagonal matrix is a diagonal matrix.

Sol.16. (a)
If the matrix is not invertible, then it is necessary that the value of the determinant of this matrix must be zero.
i.e., $\begin{vmatrix} \alpha & 2 & 2 \\ -3 & 0 & 4 \\ 1 & -1 & 1 \end{vmatrix}=0$
$\begin{pmatrix} C_2 \to C_2 + C_1 \\ C_3 \to C_3 - C_1 \end{pmatrix}$
$\begin{vmatrix} \alpha & 2+\alpha & 2-\alpha \\ -3 & -3 & 7 \\ 1 & 0 & 0 \end{vmatrix}=0$
Expanding the determinant along the third row we get
$1\begin{vmatrix} 2+\alpha & 2-\alpha \\ -3 & 7 \end{vmatrix}=0$
$\Rightarrow14+7\alpha+3(2-\alpha)=14+7a+6-3\alpha=0$
$\Rightarrow20+4\alpha=0$
$\Rightarrow4\alpha=-20\Rightarrow\alpha=-5$

Sol.17. (c)
For the matrix AB is a zero matrix. It is not necessary that either A is zero matrix or B is zero matrix.
e.g., Let $A=\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B=\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$
$\therefore AB=0$, where $A, B\neq0$

Sol.18. (b)
$\begin{bmatrix} x \\ x \\ y \end{bmatrix}+\begin{bmatrix} y \\ y \\ z \end{bmatrix}+\begin{bmatrix} z \\ 0 \\ 0 \end{bmatrix}=\begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix}$
$\Rightarrow\begin{bmatrix} x & +y & +z \\ x & +y & +0 \\ y & +z & +0 \end{bmatrix}=\begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix}$
$\Rightarrow x+y+z=10$ …(i)
$x+y=5$ …(ii)
$y+z=5$ …(iii)
From eqs. (i) and (iii),
$x+(5)=10\Rightarrow x=5$
On putting the value of x in eq. (ii), wet get
$5+y=5\Rightarrow y=0$

Sol.19. (b)
Given, $[a_{ij}]=\begin{Bmatrix} a_{ij}=0, \text{for } i=j \\ a_{ij}=k, \text{for } i=j \end{Bmatrix}$
Where k is a constant

$\therefore [a_{ij}]=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times3}$ let order 3×3

$=\begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}_{3\times3}$ =scalar matrix

Sol.20. (a)
Given that A and B are two non-singular square matrices. So, its inverse i.e., A^{-1} and B^{-1} must be exist.
We have, AB = A
(A^{-1}) operating in left side on both sides, we get
$A^{-1}(AB) = (A^{-1})(A)$
$\Rightarrow (A^{-1}A)\,B\,(A^{-1}A)$ $\quad (\therefore AA^{-1} = I \text{ and } BI = B)$
$\Rightarrow IB = I \Rightarrow B = I$ = identity matrix

Sol.21. (b)
Given, $\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 10 & -9-4+3 \\ 20 & -3-8+1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & -1 \\ 17 & -7 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$

On comparing, we get $\lambda = -7$.

Sol.22. (a)
Given that,
$A=\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, B=\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{and } C=\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$

Now, $AB = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = C$

Sol.23. (d)
We know that the product of two identity matrix are always an identity matrix, which is non-zero matrices.

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 0+0 & 1+0 \end{bmatrix}$

$=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$ = Identity matrix

The product of two non-zero matrices can sometimes be zero matrix.

$\begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \times \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$

=

$\begin{bmatrix} a+abc+bac & 0+b^2c-b^2c & 0+bc^2-bc^2 \\ a^2c+a+a^2c & -abc+0+abc & -ac^2+0+ac^2 \\ a^2b-a^2b+0 & ab^2-ab^2+0 & abc-abc+0 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ =0 = Zero matrix

So, the statements are incorrect

Sol.24. (a)
(i)Let $A = \begin{bmatrix} 1 & 2 & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{bmatrix}$

Now, |A| = 1 (2a–2b) – 2 (a–b) + 1 (2ab – 2ab)
=2a – 2b + 2b + 0 = 0
i.e., A is a singular matrix

(ii)Let $B = \begin{bmatrix} c & 2c & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{bmatrix}$

Now, |B| =c (2a–2b)–2c(a–b)+1(2b–2ab)
=2ac–2bc–2ac+2bc+0 = 0
Which is also represents a singular matrix
So, statement I is correct but statement II is incorrect.

Sol.25. (d)
Given, that, $2A=\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$

$\Rightarrow A=\frac{1}{2}\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 3/2 & 1 \end{bmatrix}$

Now, adj A =
$\begin{bmatrix} 1 & -3/2 \\ -1/2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 \\ 3/2 & 1 \end{bmatrix}$

and |A| = 1 – 3 / 4 = 1 / 4
$\therefore A^{-1} =$

$\frac{\text{adj}\,A}{|A|} = 4\begin{bmatrix} 1 & -1/2 \\ -3/2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -6 & 4 \end{bmatrix}$

Sol.26. (b)
$|\text{adj}A| = |A|^{(n-1)} = |A|^2$

Sol.27. (a)
Given matrix,
$A=\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix}$

Now, A^T =
$\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix}$

$= -\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -2 \\ -2 & 3 & 0 \end{bmatrix} = -A$

$\Rightarrow A = A^T$
So, A is skew-symmetric matrix

and $|A| = \begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}$ =0–1(0–6)+2(–3)

= 6 – 6 = 0
Since, |A| = 0 i.e., A is singular matrix
So, A cannot be an invertible matrix.

Sol.28. (b)
Given matrices,
$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}$

I. $AB=\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}_{3\times2} \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}_{2\times3}$

$=\begin{bmatrix} 1+4 & 2+2 & -4-8 \\ 2+2 & 4+1 & -8-4 \\ 1+2 & 2+1 & -4-4 \end{bmatrix}$

$=\begin{bmatrix} 5 & 4 & -12 \\ 4 & 5 & -12 \\ 3 & 3 & 8 \end{bmatrix}_{3\times3}$

II. $BA = \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}_{2\times3} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}_{3\times2}$

$=\begin{bmatrix} 1+4-4 & 2+2-4 \\ 2+2-4 & 4+1-4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times2}$

Now, we observe that B is not the right inverse of A but B is the left inverse of A.

Sol.29. (c)
Given that, A is any matrix.
Then, the product AA is defined only when A is a matrix of order m × n where, m = n i.e., A must be a square matrix.
A × A = (m × n) (m × n)
=(m × n) (n × n), if m = n
=m × n = n × n or m × n
$\therefore$ A is a square matrix.

Sol.30. (d)
We have, AB = A
$\therefore A^2$ = (AB). (BA) = A. (BA)B
=(AB)B $\quad (\because BA=B)$
=AB $\quad (\because AB=A)$
=A
Also, B^2 = (BA). (BA)
=B. (AB).A $\quad (\because AB=A)$
=(B.A).A $\quad (\because BA=B)$
=B.A
=B
Again, $(AB)^2$ = (AB).(AB)
=A.(BA)B
=(A.B).B $\quad (\because BA=B)$
=AB $\quad (\because AB=A)$

Sol.31. (a)
$\because \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

Let $B=\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ and |B| = 1

$\Rightarrow B^{-1} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} \left(\because \quad \frac{1}{|A|}\text{adj}\,A\right)$

$\therefore A=B^{-1}\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 0 & -1 \end{pmatrix}$

Sol.32. (a)
From the matrix equation AB = AC, where A, B and C are the square matrices of same order.
We can conclude B=C provided A is non singular.

Sol.33. (d)
$\because A=A'$ (A is symmetric)
$\Rightarrow \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix} = \begin{pmatrix} 4 & 2x-3 \\ x+2 & x+1 \end{pmatrix}$

$\Rightarrow 2x-3 = x+2$
$\therefore x = 5$

Sol.34. (d)
A square matrix A is said to be skew-hermitian, If $A^* = -A$
given matrix is skew-hermitian matrix.

Sol.35. (b)
$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is an elementary matrix. Since, the value of determinant of the given matrix is 1

Sol.36. (c)
We have, $A = \begin{bmatrix} 2 & 7 \\ 1 & 5 \end{bmatrix}$

$\therefore |A| = 10 - 7 = 3$

Now, $A^{-1} = \frac{1}{|A|}$ adj A

$\therefore A^{-1} = \frac{1}{3}\begin{bmatrix} 5 & -1 \\ -7 & 2 \end{bmatrix}$

$= \frac{1}{3}\begin{bmatrix} 5 & -7 \\ -1 & 2 \end{bmatrix}$

$\therefore A+3A^{-1}=\begin{bmatrix} 2 & 7 \\ 1 & 5 \end{bmatrix} + 3\times\frac{1}{3}\begin{bmatrix} 5 & -7 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = 7I$

Sol.37. (a)
$\because AX = B$ (given)
$\therefore \begin{bmatrix} p & q \\ r & s \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix}$
$\Rightarrow \begin{bmatrix} 3p+q & -4p-q \\ 3r+s & -4r-s \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix}$
$\Rightarrow 3p + q = 5$ and $-4p - q = 2$
$\Rightarrow -p = 7 \Rightarrow p = -7$
Now, $q = 5+21=26$, $3r+s= -2$
Also, $-4r - s = 1$
$\Rightarrow -r = -1$
$\Rightarrow r = 1$
and $S = -2 - 3 = -5$
$\therefore A = \begin{bmatrix} -7 & 26 \\ 1 & -5 \end{bmatrix}$

Sol.38. (a)
We have, $AB = C$
$\therefore \begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$
$\Rightarrow \begin{bmatrix} 2x+2y & -y \\ 4x & -x+y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$
$\Rightarrow \begin{bmatrix} 2x+y \\ 3x+y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$
$\Rightarrow 2x + y = 3$ and $3x + y = 2$
$\Rightarrow x = 2 - 3 = -1$
$\therefore y = 5$
Now, $A^2 =$
$\begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}^2 = \begin{bmatrix} 4 & 5 \\ -2 & -6 \end{bmatrix}\begin{bmatrix} 4 & 5 \\ -2 & -6 \end{bmatrix}$
$= \begin{bmatrix} 16-10 & 20-30 \\ -8+12 & -10+36 \end{bmatrix} = \begin{bmatrix} 6 & -10 \\ 4 & 26 \end{bmatrix}$

Sol.39. (c)
Given, $E(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$
$\therefore E(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$
and $E(\beta) = \begin{bmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{bmatrix}$
Now, $E(\alpha)\,E(\beta) =$
$\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{bmatrix}$
$= \begin{bmatrix} \cos 2.\cos\beta - \sin 2.\sin\beta & \cos 2.\sin\beta + \sin 2.\cos\beta \\ \sin 2.\cos\beta - \sin\beta.\sin 2 & -\sin 2.\sin\beta + \cos 2.\cos\beta \end{bmatrix}$
$\begin{bmatrix} \cos(\alpha+\beta) & \sin(\alpha+\beta) \\ -\sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix} = E(\alpha+\beta)$

Sol.40. (d)
$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 4 \end{bmatrix}$
So, $2x + 3A = 0$, $x = ?$
$\Rightarrow 2x + 3\begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 4 \end{bmatrix} = 0$
$\Rightarrow 2x + \begin{bmatrix} 3 & 0 & -6 \\ 6 & -9 & 12 \end{bmatrix} = 0$
$\Rightarrow 2x = \begin{bmatrix} -3 & 0 & 6 \\ -6 & 9 & -12 \end{bmatrix}$
$\Rightarrow x = \begin{bmatrix} -\frac{3}{2} & 0 & 3 \\ -3 & \frac{2}{9} & -6 \end{bmatrix}$

Sol.41. (a)
$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & x-1 & 1 \\ 2 & 7 & x-3 \end{bmatrix}$
Take $|A| = 0$
$\Rightarrow \begin{vmatrix} 1 & 3 & 2 \\ 1 & x-1 & 1 \\ 2 & 7 & x-3 \end{vmatrix} = 0$
After solving the matrix,
$\Rightarrow 1[(x-1)(x-3)-7] - 3(x-3) + 2(7-2x+2) = 0$
$\Rightarrow x^2-4x+3-7-3x+15-4x+18 = 0$
$\Rightarrow x^2-4x-4-3x-4x+33=0$
$\Rightarrow x^2-11x+29=0$
$\therefore x = \frac{11 \pm \sqrt{121-116}}{2}$
(from Sridhara Acharya formula)
Or $x = \frac{11 \pm \sqrt{5}}{2}$

Sol.42. (b)
$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$
$\Rightarrow AB = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
$\Rightarrow BA \neq O$
So, $AB \neq BA$
Thus only 2nd statement is correct.

Sol.43. (b)
$A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$
$A.A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$
$A^2 = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$
$A^2 = 2\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \Rightarrow A^2 = -2A$
$\therefore$ Statement 1 is not correct
Now,
$A^3 = A^2.A$
$= 2\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$
$A^3 = 2\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$
$A^3 = 4\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$
$A^3 = 4A$
$\therefore$ statement 2 is correct

Sol.44. (b)
Zero matrix or Null matrix

Sol.45. (d)
We have, $[x\ y\ z]\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$
$= [xa + yh + zg \quad xh + yb + zf \quad xg + yf + zc]$

Sol.46. (c)
Given, $m = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $n = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
$\therefore m\cos\theta = \begin{bmatrix} \cos\theta & 0 \\ 0 & \sin\theta \end{bmatrix}$
and $n\sin\theta = \begin{bmatrix} 0 & \sin\theta \\ -\sin\theta & 0 \end{bmatrix}$
$\therefore m\cos\theta - n\sin\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$
$\therefore$ Determinant of $(m\cos\theta - n\sin\theta)$
$= \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = 1$

Sol.47. (a)
$f(x) = \begin{vmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{vmatrix}$
$\therefore f(\theta) = \begin{vmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}$ and
$f(\theta) = \begin{vmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{vmatrix}$
$\therefore f(\theta).\,f(\phi) =$
$\begin{vmatrix} \cos\theta.\cos\theta - \sin\theta.\sin\phi + 0 & -\cos\theta.\sin\phi - \sin\theta.\cos\phi + 0 & 0+0+0 \\ \sin\theta.\cos\phi + \sin\theta\sin\phi + 0 & -\cos\theta.\sin\phi + \sin\theta.\cos\phi + 0 & 0+0+0 \\ 0 & 0 & 0+0+1 \end{vmatrix}$
$= \begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & 0 \\ \sin(\theta+\phi) & \cos(\theta+\phi) & 0 \\ 0 & 0 & 1 \end{vmatrix} = f(\theta+\phi)$
Now, determinant of $f(\theta) \times f(\phi)$
$\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & 0 \\ \sin(\theta+\phi) & \cos(\theta+\phi) & 0 \\ 0 & 0 & 1 \end{vmatrix}$
$= \cos^2(\theta+\phi) + \sin^2(\theta+\phi) = 1$
Since, determinant of $f(x) = \begin{vmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{vmatrix}$
$= \cos^2 x + \sin^2 x = 1$
$\therefore$ Determinant of $f(x)$ is an even number.

Sol.48. (d)
We have,
$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$
$\therefore AB$
$= \begin{bmatrix} 1\times2+(-1)\times(-1) & 1\times3+(-2) \\ 2\times2+3\times(-1) & 2\times3+3\times(-2) \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 0 \end{bmatrix}$
$BA = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 8 & 7 \\ -5 & -5 \end{bmatrix}$
Cleary, $AB \neq BA$
Now, $AB\,(A^{-1}B^{-1}) = AB\,(BA)^{-1}$
$[\because (AB)^{-1} = B^{-1}A^{-1}]$
$AB\,(BA)^{-1}$ is a unit matrix if and only if $AB = BA$ but $AB \neq BA$.
So, $AB\,(A^{-1}B^{-1})$ is not a unit matrix

Since, AB $\neq$ BA, therefore $(AB)^{-1}$ B^{-1}

Sol.49. (b)

$$A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$

$$\therefore A.A^T = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\alpha + \sin^2\alpha & -\cos\alpha.\sin\alpha + \cos\alpha.\sin\alpha \\ -\sin\alpha\cos\alpha + \cos\alpha\sin\alpha & \cos^2 + \sin^2\alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Sol.50. (d)

It is given that AB = C

$$\begin{bmatrix} x+y & y \\ x & x-y \end{bmatrix}\begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

$3x + 3y - 2y = 4$ or $3x + y = 4$ …(i)

and $3x - 2x + 2y = -2$ or $x + 2y = -2$ …(ii)

solving equations (i) and (ii), we get x = 2 and y = –2

$\therefore A^2 =$

$$\begin{bmatrix} 0 & -2 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 0 & -2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 0-4 & 0-8 \\ 0+8 & -4+16 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & -8 \\ 8 & 12 \end{bmatrix}$$

Sol.51. (a)

$$A^2 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta - \sin^2\theta & \cos\theta\sin\theta + \sin\theta\cos\theta \\ -\sin\theta\cos\theta - \cos\theta\sin\theta & -\sin^2\theta + \cos^2\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

$\therefore A^3 =$

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$$

Sol.52. (b)

Order of the resultant matrix

$= (1 \times 3)$ by (3×3) by $(3 \times 1) = (1 \times 1)$

Sol.53. (a)

We know that,

$$A^4 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

this is an identity matrix.

Sol.54. (a)

$$A = \begin{bmatrix} 4i-6 & 10i \\ 14i & 6+4i \end{bmatrix}$$

$$\Rightarrow kA = \frac{1}{2i}\begin{bmatrix} 4i-6 & 10i \\ 14i & 6+4i \end{bmatrix} = \begin{bmatrix} 2-\frac{3}{i} & 5 \\ 7 & 3+\frac{3}{i}+2 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\frac{3i}{i^2} & 5 \\ 7 & \frac{3i}{i^2}+2 \end{bmatrix} = \begin{bmatrix} 2+3i & 5 \\ 7 & 2-3i \end{bmatrix}$$

Sol.55. (c)

Since adj A^T – (adj A)^T

$\therefore$ adj A^T – (adj A)^T = null matrix of same order as that of A.

Sol.56. (a)

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$\therefore |A| = a\,(bc) = abc$

$$\text{Cofactor matrix of A (say B)} = \begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}$$

$$\Rightarrow \text{adj. } A = B^T = \begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}$$

$\therefore A^{-1}$

$$\frac{\text{adj.}A}{|A|} = \frac{1}{abc}\begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}$$

$$= \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$$

Sol.57. (b)

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

$$\text{Cofactor matrix of A} = \begin{bmatrix} +1 & -2 & +6 \\ +6 & +1 & -3 \\ -2 & +4 & +1 \end{bmatrix}$$

$$\text{adjoin } A = \begin{bmatrix} 1 & 6 & -2 \\ -2 & 1 & 4 \\ 6 & -3 & 1 \end{bmatrix}$$

Sol.58. (b)

$$A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 4+4 & -4-4 \\ -4-4 & 4+4 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix} = -4\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = -4A$$

Sol.59. (c)

$A_{x.x+5}$ and $B_{y.11-y}$

If AB exists then x + 5 = y or x – y = – 5

And if BA exists then 11 – y = x or x + y = 11

Solving equations, (i) and (ii), we get x = 3 and y = 8.

Sol.60. (b)

If α and β are the root of the equation $1 + x + x^2 = 0$, then $\alpha = \omega$ and $\beta = \omega^2$

$$\therefore \begin{bmatrix} 1 & \beta \\ \alpha & \alpha \end{bmatrix}\begin{bmatrix} \alpha & \beta \\ 1 & \beta \end{bmatrix} = \begin{bmatrix} 1 & \omega^2 \\ \omega & \omega \end{bmatrix}\begin{bmatrix} \omega & \omega^2 \\ 1 & \omega^2 \end{bmatrix}$$

$$= \begin{bmatrix} \omega+\omega^2 & \omega^2+\omega^4 \\ \omega^2+\omega & \omega^3+\omega^3 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix}$$

Sol.61. (a)

$A^2 - I_2 = kA$

$$\Rightarrow \begin{bmatrix} 5 & 8 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} k & 2k \\ 2k & 3k \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 8 \\ 8 & 12 \end{bmatrix} = \begin{bmatrix} k & 2k \\ 2k & 3k \end{bmatrix}$$

$\Rightarrow k = 4$

Sol.62. (a)

$(A)_{(2\times 3)} \times (B)_{(3\times 5)} = (AB)_{2\times 5}$

$\therefore$ B must be 3×5 matrix

Sol.63. (a)

In this case, A^{-1} = adj A = (co-factor A)^T

$$= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol.64. (b)

Matrix product is commutative if both are diagonal matrices of same order.

$\Rightarrow A^2 - B^2 = (A + B)(A - B)$ is not true

Next, $(A-I)(A + I) = 0$

$\Rightarrow A^2 + AI - IA - I^2 = 0$ ($\because$ AI=IA)

$\Rightarrow A^2 = I^2$ is correct.

Sol.65. (b)

B = adj A, ℓ Identity matrix, |A| = k

AB = A(adj A) = |A|ℓ = k ℓ

Sol.66. (c)

By property, statement I and III are correct.

Sol.67. (b)

$$\begin{vmatrix} 2 & 4 \\ -8 & x \end{vmatrix} = 0 \Rightarrow x = -16$$

Sol.68. (a)

$(AB)^{-1} = B^{-1}A^{-1}$

Sol.69. (a)

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

Sol.70. (b)

If $A^{-1} = A^T$, then A is orthogonal matrix.

Sol.71. (b)

Inverse of an identity matrix is an identity matrix

Sol.72. (a)

$$B = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 4 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\text{Minor} = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\text{cofactor} = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\text{Adjoint} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & -1 & 8 \end{bmatrix}$$

Sol.73. (b)

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$A^2 = I$ (Involutery matrix)

Sol.74. (a)

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}$$

$$A^3 - 2A^2 == \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix} - 2\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Sol.75. (c)

A is 3×2 order matrix and BA is 2×2 order matrix.
so AB exists but BA not exists.

Sol.76. (c)
A is 3×3 and B is 3×1 so order of AB will be 3×1

$$AB = \begin{bmatrix} ax+hy+gz \\ hx+by+fz \\ gx+fy+cz \end{bmatrix}$$

Sol.77. (c)
Order of A is 3 × 5
and order of B is 5 × 3
Then order of AB will be 3 × 3 and BA will be 5×5

Sol.78. (c)

$$A = \begin{bmatrix} 1-i & i \\ -i & 1-i \end{bmatrix}$$

$$\overline{A} = \begin{bmatrix} 1+i & -i \\ i & 1+i \end{bmatrix}$$

$$\left(\overline{A}\right)^T = \begin{bmatrix} 1+i & i \\ -i & 1+i \end{bmatrix}$$

$$\left(\overline{A}\right)^T + A = \begin{bmatrix} 1+i & i \\ -i & 1+i \end{bmatrix} + \begin{bmatrix} 1-i & i \\ -i & 1-i \end{bmatrix}$$

$$\left(\overline{A}\right)^T + A = B = \begin{bmatrix} 2 & 2i \\ -2i & 2 \end{bmatrix}$$

this matrix is hermit ion matrix because $\left(\overline{B}\right)^T = B$

Sol.79. (d)
for singular matrix, determinant of this matrix should be zero.
$-k(k-5k)+4(-k^2) = 0$
$-5k^2 + 5k^2 = 0$
for every value of k this equation is singular.

Sol.80. (b)

$$A = \begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}, B = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

AB = C

$$\begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 2x+y \\ 3x+y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

2x + y = 3..........(i)
3x + y = 2 ....(ii)
by solving above two equations
x = −1 and y = 5

then $A = \begin{bmatrix} 4 & 5 \\ -2 & -6 \end{bmatrix}$ and |A| = −14

Sol.81. (a)
I. A(adjA) = (adjA)
always correct for a square matrix
II. |adj A| = |A|
incorrect for order 3 because for order 3
$|\text{adj } A| = |A|^2$

Sol.82. (d)
order of A and B need not be same.
for example $A_{2\times3}B_{3\times2} = (AB)_{2\times2}$

Sol.83. (c)
prime numbers = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
from these 10 prime numbers 4 matrices of order (1×10), (2×5), (5×2), (10×1) are possible.

Sol.84. (a)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A \in N$$

$$\Rightarrow A.\ A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix}$$

$$\text{So } A^{100} = \begin{bmatrix} 1 & 100a \\ 0 & 1 \end{bmatrix}; A^{50} = \begin{bmatrix} 1 & 50a \\ 0 & 1 \end{bmatrix};$$

$$A^{25} = \begin{bmatrix} 1 & 25a \\ 0 & 1 \end{bmatrix}$$

So $A^{100} - A^{50} - 2A^{25}$

$$\begin{bmatrix} 1 & 100a \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 50a \\ 0 & 1 \end{bmatrix} - 2\begin{bmatrix} 1 & 25a \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 100a \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 50a \\ 0 & 1 \end{bmatrix} - 2\begin{bmatrix} 2 & 25a \\ 0 & 2 \end{bmatrix}$$

$$= -2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{os } -2I$$

Sol.85. (d)
Here we have to form distinct matrices with four entries taken from {1, 2}
So matrices formed may be of
4 × 1 or 1 × 4 or 2 × 2
So we have 3 types of matrices and each on each position either the value is 2 or 1 so there are 2 ways for each position and four entries.
So ans is = 3 × [2 × 2 × 2 × 2]= 48

Sol.86. (c)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Here (1) inverse e of A does not exist is tree because all entries are same
So |A| = 0

$\Rightarrow A^{-1}$ ie $A^{-1} = \dfrac{adjA}{|A|}$ become infinite

So (1) is correct

2.When $A^2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} or\, 2\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\text{If } A^3 = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} = 3\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

This gives yous $3A = A^3$
So, second statement ie $A^3 = A$ is incorrect and 3rd statement is correct
Option is correct.

Sol.87. (a)

$$A = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

$$\text{Adj. } A = \begin{bmatrix} -1/2 & -1 \\ -3/2 & -2 \end{bmatrix}$$

|A|=–1/2

$$A^{-1} = \frac{adjA}{|A|} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Sol.88. (d)

$$A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

(mI + nA) =

$$\left(\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} + \begin{bmatrix} 0 & 2n \\ -2n & 0 \end{bmatrix}\right)^2$$

$$= \begin{bmatrix} m & 2n \\ -2n & m \end{bmatrix}^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} m^2-4n^2 & 4mn \\ -4mn & m^2-4n^2 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$m^2 - 4n^2 = 0$
$m^2 = 4n^2$
m = 24
4mn = 2
4(2x)n = 2
$n = \frac{1}{2}$
then m = 1
$m + n = 1 + \frac{1}{2} = \frac{3}{2}$

Sol.89. (c)

1. $CA = \begin{bmatrix} m \\ -m \end{bmatrix}\begin{bmatrix} m & n \end{bmatrix} = \begin{bmatrix} m^2 & mn \\ -m^2 & -mn \end{bmatrix}$

$CB = \begin{bmatrix} m \\ -m \end{bmatrix}\begin{bmatrix} -n & -m \end{bmatrix} = \begin{bmatrix} -mn & -m^2 \\ mn & +m^2 \end{bmatrix}$

CA ≠ CB
Statement 1 is incorrect

2. $AC = [m \ \ n]\begin{bmatrix} m \\ n \end{bmatrix} = [m^2 - mn]$

$BC = [-n \ \ -m]\begin{bmatrix} n \\ -m \end{bmatrix} = [-mn + m^2]$

Statement 2 is also correct.
3. C (A + B) = CA + CB (distribution law always correct)

Sol.90. (d)
$A(\text{adj}.A) = |A|I_n$

$$A = \begin{bmatrix} 2\sin\theta & \cos\theta & 0 \\ -2\cos\theta & \sin\theta & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$|A| = 2\sin^2\theta + 2\cos^2\theta = 2$
then A (adj.A) =2I

Sol.91. (d)
For singular matrix |A|=0

$$\begin{vmatrix} 2\cos 2\theta & 2\cos 2\theta & 6 \\ 1-2\sin^2\theta & 2\cos^2\theta - 1 & 3 \\ k & 2k & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 2\cos 2\theta & 2\cos 2\theta & 6 \\ \cos 2\theta & \cos 2\theta & 3 \\ k & 2k & 1 \end{vmatrix}$$

$R_1 \to R_1 - 2R_2$

$$= \begin{vmatrix} 0 & 0 & 0 \\ \cos 2\theta & \cos 2\theta & 3 \\ k & 2k & 1 \end{vmatrix} = 0$$

Value of determinant is not depending upon the value of k.

Sol.92. (b)
Given than B = adj (A)
1.A (adj A) = (adj A) A correct
2.AB = A (adj A) = |A| I_n = scalar matrix correct.
1 and 2 only correct.

Sol.93. (b)
1.AB = 0

then A = 0 or B = 0 or $A \neq 0$ & $B \neq 0$
Incorrect statement
2.AB = I then $A = A^{-1}$
$A^{-1}A = AA^{-1} = I$
Correct statement

Sol.94. (d)
$B = A^T$
$C = A + A^T$

$$=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}+\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}2&0&0\\0&2&0\\0&0&2\end{bmatrix}$$

$|C| = 8$

Sol.95. (c)
AB = A & BA = B then $A^2 = A$ & $B^2 = B$
Both statements are correct.

Sol.96. (a)

$$P = \frac{1}{2}(A + A^T)$$

$$=\frac{1}{2}\left[\begin{pmatrix}0&\sin^2\theta&\cos^2\theta\\\cos^2\theta&0&\sin^2\theta\\\sin^2\theta&\cos^2\theta&0\end{pmatrix}+\begin{pmatrix}0&\cos^2\theta&\sin^2\theta\\\sin^2\theta&0&\cos^2\theta\\\cos^2\theta&\sin^2\theta&0\end{pmatrix}\right]$$

$$=\frac{1}{2}\begin{pmatrix}0&1&1\\1&0&1\\1&1&0\end{pmatrix}=\begin{pmatrix}0&1/2&1/2\\1/2&0&1/2\\1/2&1/2&0\end{pmatrix}$$

Sol.97. (d)

$$Q = \frac{1}{2}(A - A^T)$$

$$=\frac{1}{2}\left[\begin{pmatrix}0&\sin^2\theta&\cos^2\theta\\\cos^2\theta&0&\sin^2\theta\\\sin^2\theta&\cos^2\theta&0\end{pmatrix}-\begin{pmatrix}0&\cos^2\theta&\sin^2\theta\\\sin^2\theta&0&\cos^2\theta\\\cos^2\theta&\sin^2\theta&0\end{pmatrix}\right]$$

$$=\frac{1}{2}\begin{pmatrix}0&\sin^2\theta-\cos^2\theta&\cos^2\theta-\sin^2\theta\\\cos^2\theta-\sin^2\theta&0&\sin^2\theta-\cos^2\theta\\\sin^2\theta-\cos^2\theta&\cos^2\theta-\sin^2\theta&0\end{pmatrix}$$

$$=\frac{1}{2}\begin{pmatrix}0&-\cos2\theta&\cos2\theta\\\cos2\theta&0&-\cos2\theta\\-\cos2\theta&\cos2\theta&0\end{pmatrix}$$

$$=\frac{1}{2}\cos2\theta\begin{pmatrix}0&-1&1\\1&0&-1\\-1&1&0\end{pmatrix}$$

Sol.98. (a)

$$A = \begin{pmatrix}0&\sin^2\theta&\cos^2\theta\\\cos^2\theta&0&\sin^2\theta\\\sin^2\theta&\cos^2\theta&0\end{pmatrix}$$

$|A| = \sin^6\theta + \cos^6\theta$.
And its min value exits at 45°
So Min|A| = ¼

Sol.99. (d)

$$A = \begin{pmatrix}1&0&0\\0&\cos\theta&\sin\theta\\0&\sin\theta&-\cos\theta\end{pmatrix}$$

$$\text{adj}A = \begin{pmatrix}-1&0&0\\0&-\cos\theta&-\sin\theta\\0&-\sin\theta&\cos\theta\end{pmatrix}$$

$$A^{-1} = \begin{pmatrix}1&0&0\\0&\cos\theta&\sin\theta\\0&\sin\theta&-\cos\theta\end{pmatrix}$$

, then which of the following are correct?
1. A + adjA is a null matrix (correct)
2. A^{-1} + adjA is null matrix (correct)
3. $A - A^{-1}$ is a null matrix (correct)

Sol.100. (d)
If X is a matrix of order 3 × 3, Y is a matrix of order 2 × 3 and Z is a matrix of order 3 × 2, then which of the following are correct?
1. (ZY) is a 3 × 3 matrix,
then (ZY)X is a 3 × 3 matrix so matrix having 9 entries
2. (XZ) is a 3 × 2 matrix ,
Then Y(XZ) is a 2 × 2 matrix so having 4 entries
3.X(YZ) is not defined because number of columns in X matrix is not equal to number of rows in YZ matrix
All above statements are correct

Sol.101. (c)
A and B be symmetric matrices
(AB – BA)' = (AB)' – (BA)'
(AB)' – (BA)' = B'A' – A' B' = BA – AB
(AB – BA)' = BA – AB =
(AB – BA)' = – (AB – BA)
So (AB – BA) is a skew symmetric matrix and for any skew symmetric matrix below given statements are always correct
1.Its diagonal entries are equal but nonzero
2.The sum of its non-diagonal entries is zero

Sol.102. (c)
1. AB = AC
$\Rightarrow A^{-1}AB = A^{-1}AC$
$\Rightarrow B = C$
2. If BX = CX
If X is a column matrix then
B = C
Both statements are correct

Sol.103. (b)
X_1 and X_2 be its two distinct solutions of AX = B
So $AX_1 = B$ ……….(i)
And $AX_2 = B$ ………….(ii)
$aX_1 + bX_2$ is a solution of AX = B
so $A(aX_1 + bX_2) = B$
by (i) and (ii)
aB + bB = B
a + b = 1

Sol.104. (a)

$$AA' = \begin{bmatrix}1\\2\\3\end{bmatrix}\begin{bmatrix}1&2&3\end{bmatrix}=\begin{bmatrix}1&2&3\\2&4&6\\3&6&9\end{bmatrix}$$

(I+AA') =

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}+\begin{bmatrix}1&2&3\\2&4&6\\3&6&9\end{bmatrix}=\begin{bmatrix}2&2&3\\2&5&6\\3&6&10\end{bmatrix}$$

|I+AA'|= 28 – 4 – 9 = 15

Sol.105. (a)

$$A = \begin{bmatrix}0&3&4\\-3&0&5\\-4&-5&0\end{bmatrix},$$

$\Rightarrow (A^2)' = (AA)' = A'A'$
[Given matrix is a skew symmetric
So A'= –A]
$= (-A)(-A) = A^2$
So A^2 is a symmetric matrix.
Determinant of an odd ordered skew symmetric matrix is zero
So |A| = 0 , then $|A^n| = 0$

Sol.106. (b)
If $A = \begin{bmatrix}2&0&0\\0&3&0\\0&0&4\end{bmatrix}$, then which one of the following statements are correct?
1.$|A| \neq 0$, so A and A^n is not a singular matrix
2.if we multiply diagonal matrix to any other diagonal matrix then result will be a diagonal matrix.
3.Diagonal matrix is always a symmetric matrix
Only statement 2 and 3 are correct.

Sol.107. (d)

$$A = \begin{vmatrix}2&-1&0\\-1&3&0\\1&0&1\end{vmatrix}$$

$\Rightarrow |A| = 5$

$$\Rightarrow |\text{adj}(\text{adj}A)| = |A|^{(n-1)^2} = 5^4 = 625$$

Sol.108. (a)
If $A = A^2 = A^3 = A^n$ = unit matrix = $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$,

$23A^2 - 19A^2 - 4A$

$$= 23\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}-19\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}-4\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$=\begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}$$

Sol.109. (d)
Number of matrices of order 2 × 2 are 4×4×4×4 = 256
Number of matrices of order 1 × 4 are 4×4×4×4 = 256
Number of matrices of order 4 × 1 are 4×4×4×4 = 256
Toctal matrices = 768

Sol.110. (c)

$$A = \begin{bmatrix}\sin2\theta&2\sin^2\theta-1&0\\\cos2\theta&2\sin\theta\cos\theta&0\\0&0&1\end{bmatrix}$$

$$A = \begin{bmatrix}\sin2\theta&-\cos2\theta&0\\\cos2\theta&\sin2\theta&0\\0&0&1\end{bmatrix}$$

det(A) = 1
so A^{-1} = adj(A)
diagonals are not zero so matrix is not a skew symmetric matrix.
and $A^{-1} = A^T$

statement 1 and 3 are correct.

Sol.111. (b)

adj(AB) = (adjB)(adjA)

so statement 1 and 2 are incorrect.

(A)adj(A) = $|A|I_n$

Statement 3 is correct.

Sol.112. (a)

from properties of adjoint

$A(adjA^T) = A(adjA)^T$

statement 1 is correct.

$A^2 = A$

$A(A - I) = 0$

$|A| = 0$ or $|A| = 1$

matrix is non singular given in question so |A| = 1, and orthogonal matrix of |A| = 1 is always a identity matrix.

statement 2 is correct.

$A^3 = A$

$A(A^2 - I) = 0$

$|A| = 0$ or $|A| = +1$ or -1

matrix is non singular given in question so |A| = +1 or – 1

so always unit matrix is not correct.

statement 3 is wrong.

Sol.113. (a)

diagonals of skew symmetric matrix is always zero. so statement 1 and 2 are correct.

orthogonal matrix $A^{-1} = A^T$ or $AA^T = I$

skew symmetric matrices can never be orthogonal matrix.

Sol.114. (d)

$$A=\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & -1 & 1\end{bmatrix}$$

for above matrix |A| = 1

$A(adjA) = |A|\ I_n$

$$A(adjA)=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

Sol.115. (a)

$$A=\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & -1 & 1\end{bmatrix}$$

for above matrix |A| = 1

$$A^{-1} = \frac{1}{|A|}\text{ adj A} = \text{adj A} = \begin{bmatrix}1 & -1 & 0\\ -2 & 3 & -4\\ -2 & 3 & -3\end{bmatrix}$$

Sol.116. (c)

$$P=\begin{bmatrix}0 & c & -b\\ -c & 0 & a\\ b & -a & 0\end{bmatrix}$$

and

$$Q=\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}$$

$$PQ=\begin{bmatrix}0 & c & -b\\ -c & 0 & a\\ b & -a & 0\end{bmatrix}\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$QP=\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}\begin{bmatrix}0 & c & -b\\ -c & 0 & a\\ b & -a & 0\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

PQ = QP = Null matrix

statement I and III are correct.

Sol.117. (c)

If X is a matrix of order 3 × 3, Y is a matrix of order 2 × 3 and Z is a matrix of order 3 × 2, then which of the following are correct?

1. (ZY) is a 3 × 3 matrix,
then (ZY)X is a 3 × 3 matrix
2. (XZ) is a 3 × 2 matrix ,
then Y(XZ) is a 2 × 2 matrix
3. X(YZ) is not defined because number of columns in X matrix is not equal to number of rows in YZ matrix

All above statements are correct

Sol.118. (c)

AB = 0

If any one matrix is non-singular than second matrix should be a null matrix.

Sol.119. (d)

$$\begin{bmatrix}x+4+7 & 2x+5+8 & 3x+6+9\end{bmatrix}\begin{bmatrix}1\\ 1\\ x\end{bmatrix}=\begin{bmatrix}45\end{bmatrix}$$

$$\begin{bmatrix}x+11+2x+13+3x^2+15x\end{bmatrix}=\begin{bmatrix}45\end{bmatrix}$$

$3x^2 + 18x - 21 = 0$

$x^2 + 6x - 7 = 0$

$(x + 7)(x - 1) = 0$

$x = 1, -7$

Sol.120. (b)

$$A=\begin{bmatrix}y & z & x\\ z & x & y\\ x & y & z\end{bmatrix}$$

If A is orthogonal than $A.A^T = I$

$$AA'=\begin{bmatrix}y & z & x\\ z & x & y\\ x & y & z\end{bmatrix}\begin{bmatrix}y & z & x\\ z & x & y\\ x & y & z\end{bmatrix}=I$$

$$=\begin{bmatrix}x^2+y^2+z^2 & xy+yz+zx & xy+yz+zx\\ xy+yz+zx & x^2+y^2+z^2 & xy+yz+zx\\ xy+yz+zx & xy+yz+zx & x^2+y^2+z^2\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$x^2 + y^2 + z^2 = 1$

Sol.121. (c)

$$\left|M^{-1}\right|=\frac{1}{|M|}$$

2nd property is wrong, others are correct

Sol.122. (d)

$$f(\theta)=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$$

$$f(\pi)=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$$

$$\{f(\pi)\}^2=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

Sol.123. (d)

$$A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$$

$$A^2-4A=\begin{bmatrix}9 & 8 & 8\\ 8 & 9 & 8\\ 8 & 8 & 9\end{bmatrix}-\begin{bmatrix}4 & 8 & 8\\ 8 & 4 & 8\\ 8 & 8 & 4\end{bmatrix}=\begin{bmatrix}5 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{bmatrix}$$

$= 5I_3$

Sol.124. (b)

$$A=\begin{bmatrix}y & z & x\\ z & x & y\\ x & y & z\end{bmatrix}$$

If A is orthogonal than $A.A^T = I$

$A^2 = I$

Sol.125. (a)

I. $(AB)^{-1} = B^{-1}A^{-1}$

II. $(BA)(AB)^{-1} = (BA)(B)^{-1}(A)^{-1}$

III. $(AB)^T = B^T A^T$

So all are incorrect.